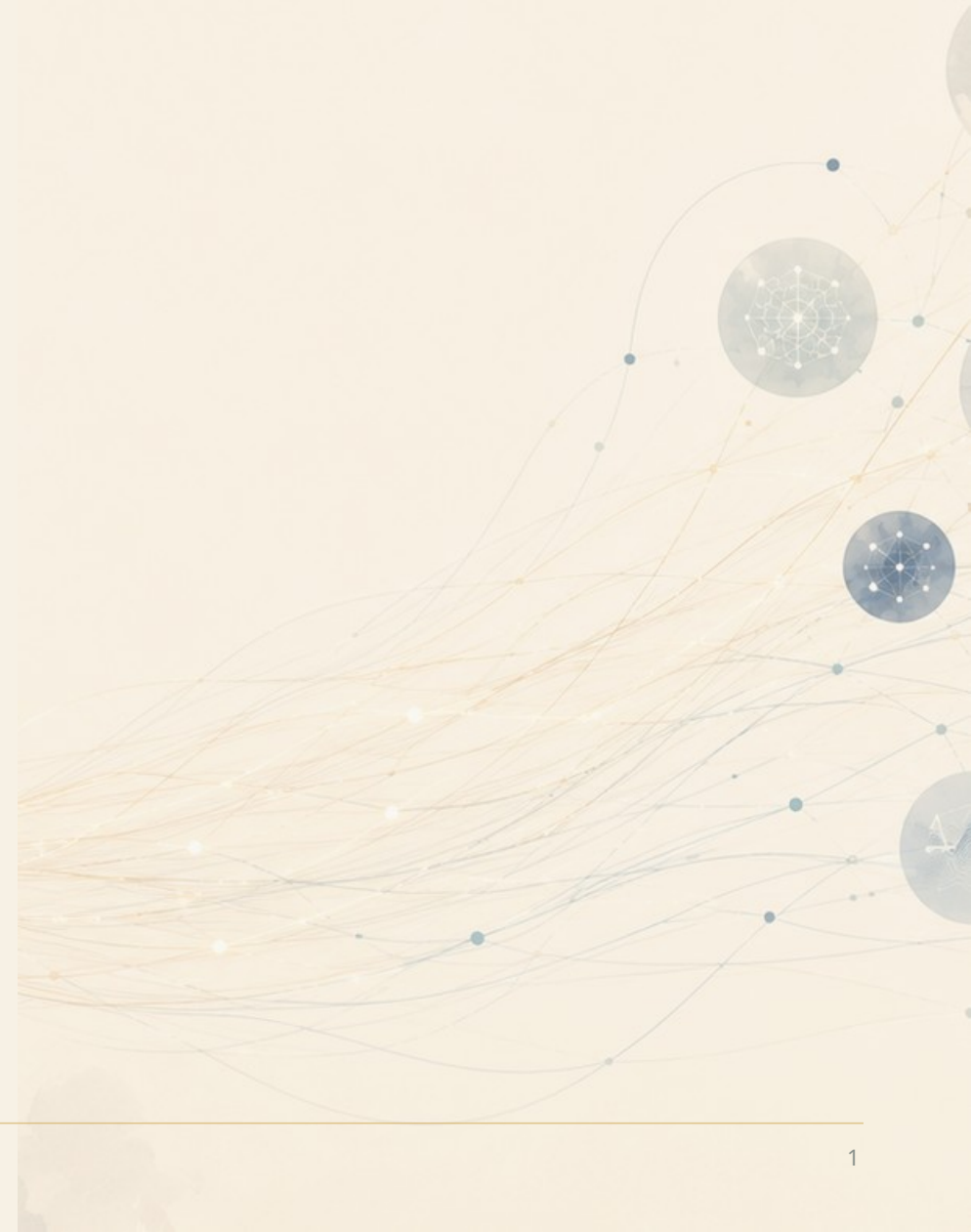


The Future of AI May Feel Like One Relationship

A healthcare-focused forecast about identity, context, memory, authority, and responsibility.

HCOS™ Insight + Architectural Forecast

D-002 Human-Centered AI



CME Activity Information

Replace bracketed fields with provider-approved language before publication.

Accredited provider	[Insert accredited provider / joint provider]
Credit designation	[Insert approved CME / CE credit language]
Release / expiration dates	[Insert release, review, and expiration dates]
Activity format	[Live / enduring material / blended]
Credit claiming	[Insert CE21 or LMS instructions]



Faculty, Planning Committee & Disclosure

Insert required disclosure and mitigation language.

Faculty

Carey Jones, PharmD
[Insert title, organization, and credentials as appropriate]

Disclosures

Relevant financial relationships: [Insert none or details]
Mitigation: [Insert approved mitigation statement]

Support

Commercial support: [Insert statement]
AI-assisted development: [Insert transparency statement if used]



Learning Objectives

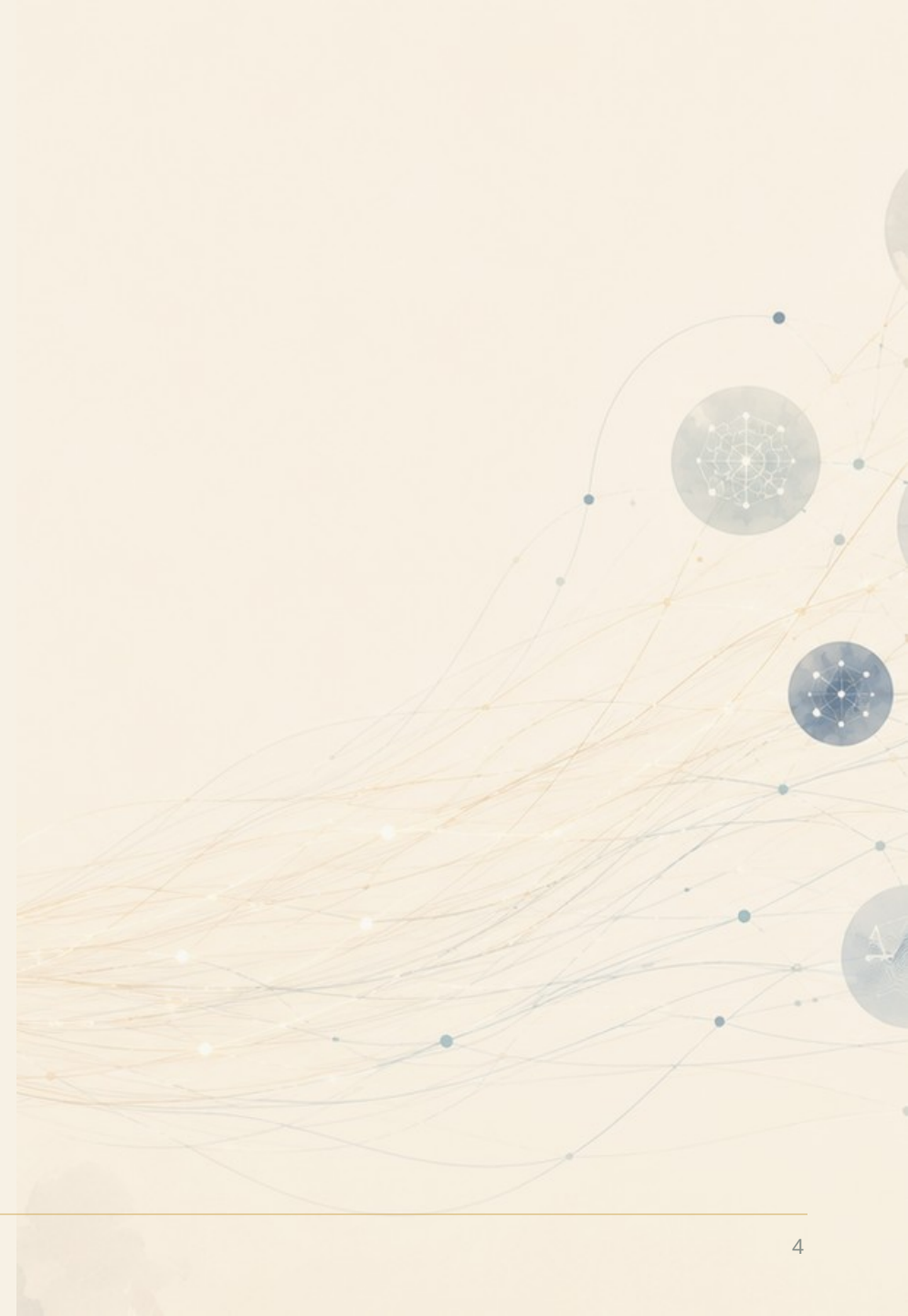
After participating in this activity, learners should be able to:

Differentiate a continuous AI experience from the distributed systems operating underneath it.

Describe why healthcare AI requires context-bound identity, delegated authority, and governed memory.

Identify boundary risks when AI moves across patient, clinician, organizational, payer, and family contexts.

Apply a practical stewardship lens before AI-supported work moves from information to action.



Educational Scope & Disclaimer

This activity provides conceptual education for healthcare professionals.

This is not legal, regulatory, cybersecurity, privacy, billing, or clinical implementation advice.

Examples are illustrative and must be adapted to local policy, workflow, governance, and patient needs.

AI tools should not make patient-specific decisions without appropriate validation, review, and accountability.

Healthcare organizations remain responsible for safety, privacy, fairness, documentation, and professional judgment.



Why this matters in healthcare

Healthcare already runs on many identities, roles, systems, and permissions.



“Continuity is helpful only when boundaries remain visible.”

In healthcare, trust depends on knowing who is acting, in what role, with what information, and under whose authority.

HCOS™ means Human-Centered Operating Systems

A way to examine technology as part of a larger human system.

HCOS asks:

Does this system support people, preserve dignity, reduce unnecessary burden, and keep responsibility visible?

Technology

What the tool can do

Workflow

How work actually moves

Authority

Who may act

Accountability

Who remains responsible

People may stop asking which application to open

They may start asking what outcome they want.

Today

Email
Calendar
Maps
Banking
EHR / portal



Tomorrow

**"Help me plan
my daughter's
birthday."**

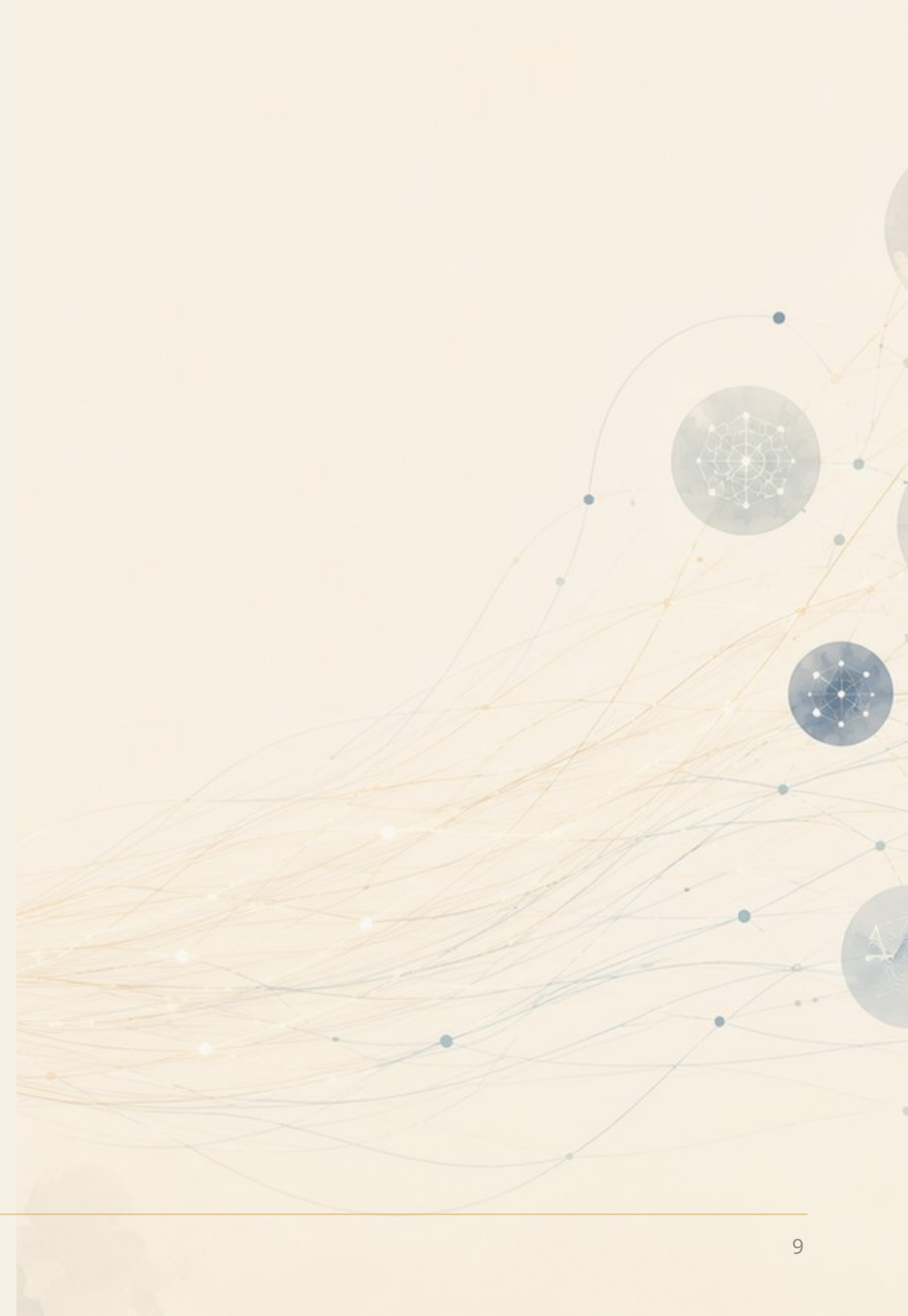


One request may require many systems

The person experiences one request. The system performs coordinated work.

- 01 Check calendars
- 02 Find venues
- 03 Draft invitations
- 04 Coordinate family
- 05 Update budget
- 06 Set reminders

“The experience may be simple. The architecture is not.”



A healthcare AI request is rarely one task

A single goal may cross clinical, operational, financial, and patient-support systems.

“Help this patient start therapy safely and on time.”

Clinical criteria and documentation
Benefit coverage and prior authorization
Lab and safety monitoring
Patient education and adherence support
Scheduling, pharmacy, and follow-up

Key question: what may the AI see, remember, decide, draft, send, or do?



Which safeguard matters most?

A healthcare AI coordinates follow-up across scheduling, pharmacy, and insurance systems.

What is the most important architectural safeguard?

- A. Use the same AI model for every task
- B. Allow all systems to access the complete patient record
- C. Limit each capability to the identity, context, and authority needed for its task**
- D. Remove human review to improve efficiency



Correct answer: C

Limit each capability to the identity, context, and authority needed for its task.

"A continuous experience should not mean unlimited access."

Use the minimum information needed for the task.
Separate retrieval, drafting, communication, submission, and decision-making.
Preserve human review where the action becomes consequential.

Teaching point: continuity should reduce burden without collapsing permission boundaries.



AI actions exist at different levels of consequence

Risk increases as AI moves from information toward action.



Example: explain requirements → identify missing documentation → draft a message → send it → submit an authorization → complete a transaction

The more the AI acts, the more explicit the identity, permission, review, and audit trail must become.

Authority should be assigned by task

Do not grant the whole workflow one broad permission.

AI role	Required human boundary
Retrieve + summarize	Clinician verifies
Draft documentation	Professional approves
Contact patient	Authorized sender/reviewer
Submit authorization	Explicit delegation + audit
Recommend therapy change	Licensed professional decides

“The task defines the authority.”

The five questions before AI acts

Use these before AI-supported work moves from information to action.

- | | | |
|---|------------------------------------|--|
| 1 | Who is acting? | Patient, clinician, pharmacy, payer, or organization? |
| 2 | What is the purpose? | What specific outcome is being pursued? |
| 3 | What information is needed? | Is access limited to this task? |
| 4 | What may the AI do? | Retrieve, summarize, draft, send, submit, modify, or decide? |
| 5 | What review is required? | Who confirms the action, and who remains accountable? |



Drafting is not the same as sending

An AI drafts a patient message about missing lab work and then sends it automatically.

What is the most important governance concern?

- A. The AI should use more conversational language
- B. Sending is a delegated action that requires defined authority and review**
- C. The message should be shorter
- D. The AI should remember the message forever

Correct answer: B

Sending is a delegated action that requires defined authority and review.

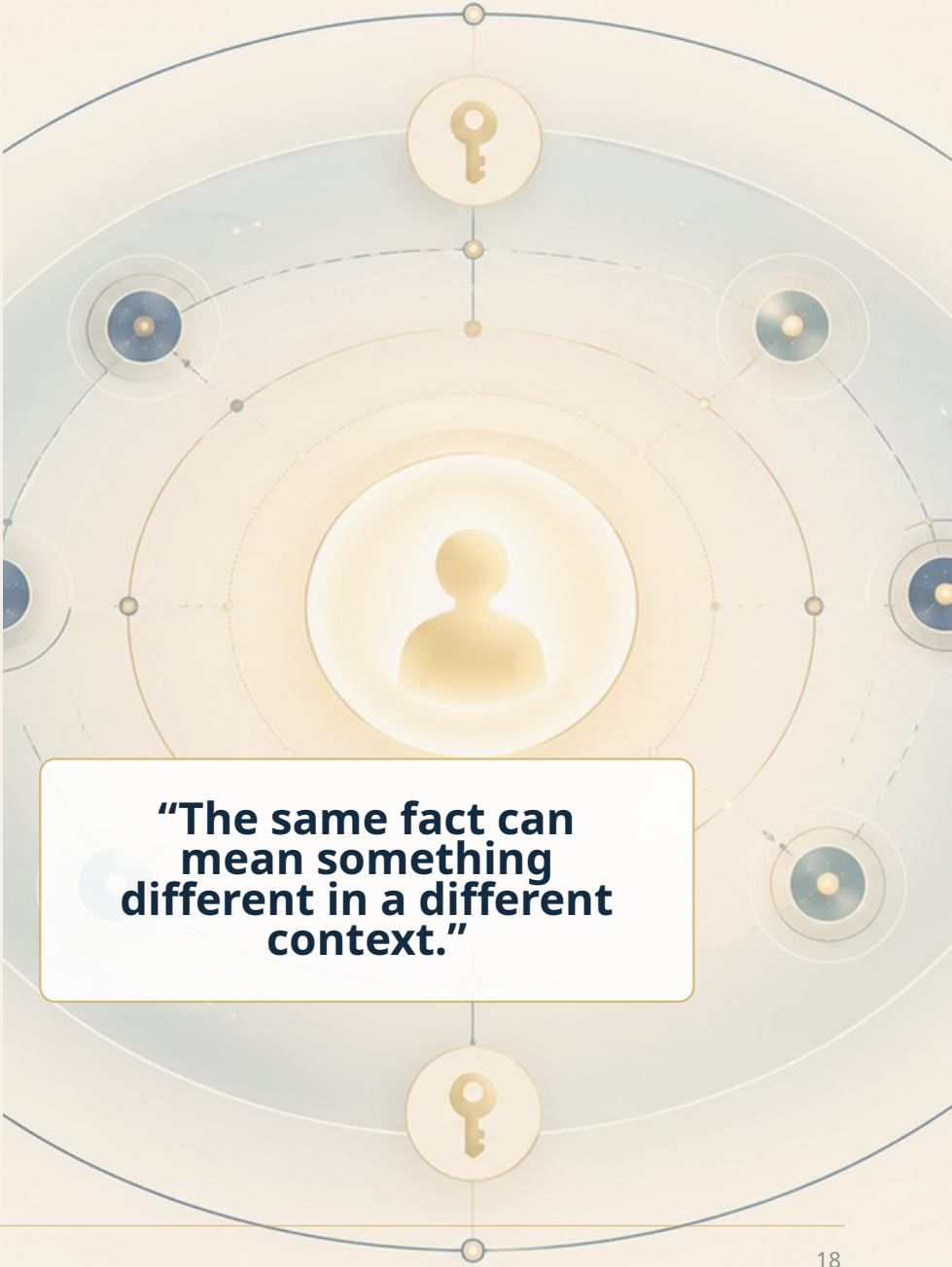
"A draft prepares communication. Sending performs communication."

Define who the AI is representing.
Separate draft generation from message release.
Match review requirements to consequence, uncertainty, and patient impact.

Some AI failures are boundary failures

They may look like reasoning errors, but the deeper issue is identity, context, or authority.

A caregiver preference is mistaken for the patient's preference.
Old information is treated as current.
A draft recommendation is interpreted as an approved clinical decision.
An AI acting for the organization appears to be acting for the clinician.
Information collected for care coordination is reused for another purpose.



"The same fact can mean something different in a different context."

Memory should preserve meaning, not just data

Every remembered item should carry boundaries.

Source

Where did this come from?

Context

When and where was it true?

Consent

Was persistence appropriate?

Correction

Can it be fixed?

Expiration

When should it be reconfirmed?

Audit

Where did it influence action?

““The patient prefers communication through their spouse.””

This may be a preference, a temporary consent, a caregiver note, or a boundary violation—depending on context.

Match review to consequence

Not every AI action requires the same human checkpoint.

Low consequence

Formatting
Scheduling suggestions
Educational summaries

Moderate consequence

Draft patient messages
Identify missing documentation
Route information

High consequence

Submit authorizations
Modify orders
Initiate treatment-related action

Review should be matched to consequence, uncertainty, reversibility, and professional responsibility.



Where is the boundary failure?

An AI reviews a record, drafts an authorization, messages the prescriber, and submits the request before pharmacist review.

What is the primary governance problem?

- A. The AI used too many systems
- B. The AI exceeded delegated authority before the review threshold was met**
- C. The request involved insurance
- D. The workflow included pharmacy



Correct answer: B

The AI exceeded delegated authority before the review threshold was met.

“Capability is not permission.”

The AI may assist with retrieval and drafting.
Submission is a delegated transaction, not just information support.
A responsible human or approved governance pathway must define when action can occur.



Start with one real AI-enabled workflow

Map the work before expanding the authority.

Map every person, system, and organization involved.
Separate retrieval, drafting, communication, decision, and action.
Define the information each task actually requires.
Set authority and review thresholds before deployment.
Name the accountable human or organizational owner.
Create correction, escalation, and audit pathways.
Evaluate whether AI reduces or transfers human burden.



Before AI acts, ask:

Who is it representing?

What context is active?

What information is necessary?

What authority has been delegated?

What review is required?

Who remains responsible?

“One trusted experience can still require many carefully governed systems.”

Post-test + evaluation

Complete credit-claiming steps in CE21 or designated LMS. [Insert official provider instructions.]

References: NIST AI RMF 1.0 (2023); NIST Generative AI Profile (2024); WHO Ethics and Governance of AI for Health (2021); ONC HTI-1 / Decision Support Intervention resources (2024–2026).